

Communicate:

T-test for the slope of a least-squares regression line

β = the true slope of the regression line relating daily AI usage (in hours) to GPA for all students in Doral Academy.

$$H_0: \beta = 0 \quad H_a: \beta < 0$$

Conditions:

◦ Linear: scatterplot shows a roughly linear negative pattern.

◦ Independent: 50 < 10% of all the high school students in Doral Academy

◦ Normality: residuals look approximately normal, no strong skewness nor outliers

◦ Equal variance: No funnel shape

◦ Random: we concluded a stratified random sample

Since all conditions are met, we can proceed!

Calculations:

$$t = \frac{b - 0}{SE_b} = \frac{-0.4735}{0.0989} \approx -4.788$$

$$df = n - 2 = 50 - 2 = 48$$

$$P(T_{48} \leq -4.79) = P(T_{48} \geq 4.79)$$

$$t.cdf(4.79, 99999, 48) \approx 0.0000083$$

$$p \approx 0.0000083$$

Conclude:

Since our p-value is 0.0000083 is less than our significance level of 0.05, we reject the null hypothesis. Therefore, there is strong evidence statistically that there is a negative linear relationship between AI usage and GPA.